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Microgravity flammability limits of ETFE insulated wires exposed to external radiation

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Abstract

The present work studied the normal gravity (1 g) and microgravity (μ g) flame spread limits (LOC) of ETFE insulated copper wires exposed to an external radiant flux. Experiments with sample wires of a 0.50 mm copper core and 0.30 mm ETFE insulation thickness were conducted in oxygen concentrations ranging from 20% to 32% and external radiant fluxes from 0 to 25 kW/m². Microgravity experiments conducted in parabolic flights showed that μ g reduced the Limiting Oxygen Index of the material. The addition of an external radiant flux further extends the Limiting Oxygen Concentration (LOC) for flame spread over ETFE insulated wires. Microgravity reduced heat losses and allowed the flame to propagate in lower oxygen concentrations. The addition of an external radiant flux further compensates for lower flame temperatures in reduced oxygen concentrations and further extends the LOC of the material. Limiting Oxygen Index (LOI) results obtained with ETFE were also compared to available results with PE and show that μ g conditions have a larger impact in ETFE than PE. The results of this work are relevant given that the flammability of materials is routinely tested without considering the effects of environmental variables and according to the results presented in here may not be indicative of the absolute flammability limits. © 2014 The Combustion Institute. Published by Elsevier Inc. All rights reserved.

Keywords: Microgravity; Ethylene-tetrafluoro-ethylene (ETFE); Limiting Oxygen Concentration (LOC); External radiation

Abbreviations: FR, fire resistant; PE, polyethylene; HRP, heat release parameter; HRR, heat release rate; LOC, Limiting Oxygen Concentration; LOI, Limiting Oxygen Index; MOC, Maximum Oxygen Concentration; ETFE, ethylene-tetrafluoro-ethylene; PMMA, poly methylmethacrylate.

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1. Introduction

The risk of a fire during a space mission has motivated ample research in spacecraft fire safety. Over the years researchers have studied the differences between flame spread in normal (1 g) and microgravity (μ g) conditions using different fuels such as plastics, paper sheets and electrical cables. Findings from these studies have shown that in low flow velocities typical of spacecraft ventilation

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systems, μg results in higher flame spread rates, lower flammability limits, and different flame extinction mechanisms when compared to 1 g [1–10].

Electrical cables and harnesses have been identified as a possible source of fires in a spacecraft and their combustion in microgravity has received significant attention. Kikuchi et al. [6] studied the effects of wire temperature and size, oxygen concentration, ambient pressure, and carrier gas on the flame spread over ETFE insulated wires. In these experiments the authors found that the combination of μg and preheating of the wire core resulted in an increase in the flame spread rate in comparison to 1 g. The same authors also found that smaller wire sizes resulted in higher spread rates, and that as wire size increased the difference between microgravity and normal gravity flame spread rates decreased. Fujita et al. [11] studied the effect of oxidizer flow speed in flame spread over PE insulated wires. One key finding was that in μg , flame spread rate had a maximum at flow velocities around 0.10 m/s, which are similar to those induced by air ventilation systems inside a spacecraft. This agreed with observations by Olson and collaborators [1,2], who also found that flow velocities ranging from 0.06 to 0.10 m/s resulted in the highest flame spread rate over thin cellulosic fuels.

Fujita et al. [8] studied the ignition delay and ignition limits of PE insulated nichrome wires subject to short-term excess current. The results showed that the minimum current required for ignition and ignition delay times decreased in microgravity. Expanding on the ignition of short-term overloaded wires, Takano et al. [9] investigated the LOC for the ignition of PE insulated nichrome wires. In this study, the LOC was defined as the oxygen concentration below which ignition does not occur given a certain current supply. The results showed a reduction in the LOC in μg . Takahashi et al. [10] also observed that microgravity results in lower LOC for flame spread over PE insulated copper and nichrome wires, which was related to the elimination of natural convection and the resulting increased heating of the unburned insulation.

As microgravity fire safety research has shifted to the study of ignition and extinction limits, the types of fuels have also changed. Except a few studies, the majority of microgravity flame spread research has been conducted using materials that help the basic understanding of the problem, like for example thin cellulosic paper, PMMA sheets or PE insulated wires. Still, microgravity flame spread experiments with more practical materials used in space applications are less common. Development of a next generation of space exploration vehicles with elevated oxygen concentrations and reduced ambient pressure cabins [12] has brought attention to the flammability behavior

of FR materials under these cabin atmospheres [13–17]. Fire resistant materials have a LOI greater than 21%. The LOI is defined as the minimum oxygen concentration that supports a candle like flame [18]. Some of these studies have shown that although FR materials are not flammable in normal atmospheric conditions, they can become flammable in elevated oxygen concentrations, reduced ambient pressure, or microgravity. Also, studies on the effect of an external radiant flux on the flammability of materials have shown that an external radiant flux extends the flammability limits of materials below the LOI [19–21].

Fire resistant wire insulation materials are extensively used in spacecraft. One material that has been used in insulation and wire harnesses is ETFE. In standard atmospheric conditions ETFE has a LOI of 30% [22]. Based on the work described above it is possible that the flammability of ETFE may be extended in microgravity and under external heating, and constitutes the objective of the present work. This objective is accomplished by studying the LOI and LOC of ETFE insulated wires subject to an external radiant flux in both 1 g and μg . In addition, the ETFE insulated wire LOI results are also compared to results with PE insulated wires obtained by Takahashi et al. [10]. The comparison helps understanding whether the effect of μg in the LOC can be predicted using non fire-resistant materials such as PMMA, PE, etc.

2. Experiment configuration

A schematic of the experimental apparatus is shown in Fig. 1. The apparatus and supporting equipment is designed to fit in an experimental flight rack for microgravity experiments. The main component of the apparatus is a 60 mm in diameter and 250 mm long flow duct made of Pyrex. The ETFE insulated wire is placed at the centerline of the duct and it is fed via a spool system located at both ends of the tube. The combustion products are filtered with an air filter located downstream of the duct. A suction fan (Sanyo Denki 9GV0612P1H031) attached to the air filter is used to induce a recirculating airflow throughout flow duct. Honeycomb flow straighteners placed at both ends of the tube are used to ensure flow uniformity. A constant airflow speed of 0.12 m/s through the flow duct was used for all experiments. This flow velocity simulates typical airflow speeds found in spacecraft ventilation systems. Also, it has been found that opposed flame spread rates have maximum values around these flow rates [1,2,8,10,11].

Two 300 mm long halogen heaters (Ushio UH-USC-CL300) with a peak wavelength emission of 1.2 microns are mounted directly on top and

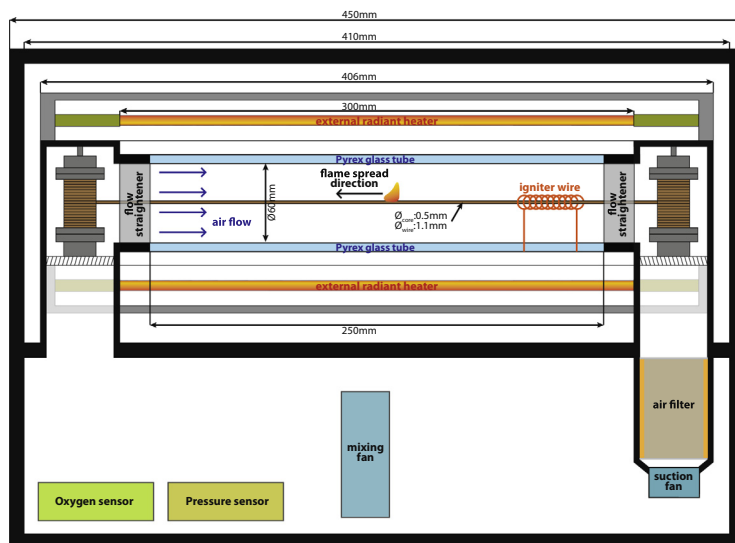


Fig. 1. Schematic of flow duct used for the experiments.

below the flow duct. The heaters are contained in an isolated volume that limits exposure to toxic and corrosive ETFE combustion products. The heaters are controlled using a transformer and provide a maximum radiant flux of 25 kW/m^2 to the wire sample. A heat flux sensor (International Thermal Instrument Company ITI-HT-50) is used to verify a uniform heat flux distribution along the wire surface. ETFE insulated copper wire samples (Sugita Densen Co., Ltd.) with a core diameter of 0.50 mm and an insulation thickness of 0.30 mm are used in all experiments. Ignition of the wire insulation is accomplished using a 0.5 mm thick Kanthal wire coil with a 8 mm diameter. The coil is concentric with the wire and is placed downstream end of the wire. A programmable logic controller (Mitsubishi MELSEC FX2N-16MR) is used to control a constant current power supply (Takasago EX-375L2) that energizes the igniter coil for 12 s with a maximum power of 92 W (14.7 V, 6.3 A). In microgravity experiments, the igniter was energized 10 s before the beginning of the μg period to reduce the influence of the igniter during μg flame spread.

A pressure sensor (Nagano Keiki KP15–17G) and oxygen sensor (Jikco JKO-25LII) are used to monitor pressure and oxygen concentration in the chamber. The maximum change in the oxygen concentration observed during an experiment was 1.0% O_2 . One type-T thermocouple is used to measure air temperature in the chamber. The gas inside the chamber consists of mixtures of oxygen and nitrogen. The desired oxygen concentration inside the chamber is set using oxygen partial pressures with a total chamber pressure of 100 kPa. Once a test starts the external radiant heaters are turned on and allowed to irradiate the sample wire. The igniter is turned on after a

30 s preheating time. Microgravity experiments are conducted in parabolic flights onboard a MU-300 aircraft operated by Diamond Air Service (DAS) in Nagoya, Japan. Each parabola provides about 20 s of 10^{-2} g . Chamber pressure, oxygen concentration, air temperature, and acceleration in the case of microgravity experiments, were recorded using a data acquisition system (Graphtec GL220 midi LOG- GER dual).

Experiments consisted in determining whether opposed flame spread occurred along the wire insulation for a given set of ambient oxygen concentration and external radiant flux conditions. The oxygen concentration resulting in a change from No Propagation to Propagation was defined as the LOC. The LOC values as a function of the applied external radiant flux are used to produce 1 g and μg LOC boundaries.

3. Results

3.1. Normal gravity (1 g)

Figure 2 presents a typical ETFE insulation flame during experiments in normal gravity. Normally flames wrapped around the cable insulation and had upward elongated shapes as a result of buoyancy. In oxygen concentrations near the LOC the unburned ETFE insulation at the leading edge of the flame softened and started to accumulate. When the accumulation was large enough it created a droplet that eventually dripped onto the Pyrex tube. An insulation residue on the surface of the copper wire was observed in most tests.

The experimental results are presented in Fig. 3 in the form of a Propagation/No-Propagation boundary (LOC) as a function of the radiant flux

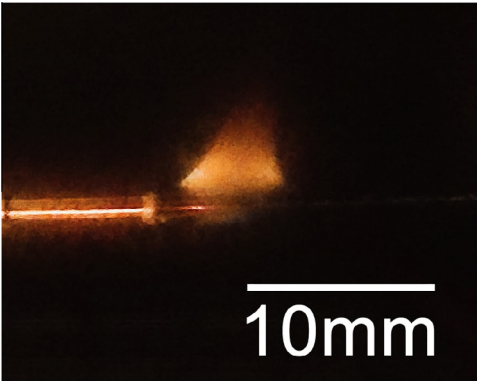


Fig. 2. ETFE opposed flame spread in normal gravity. Test conditions were 26.3% O₂ and $\dot{q}''_{ext} = 24.3 \text{ kW/m}^2$.

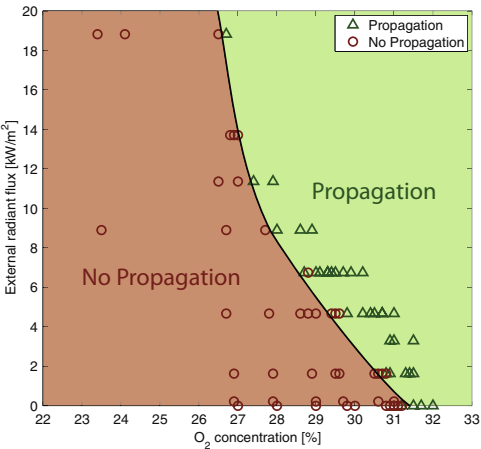


Fig. 3. Normal gravity flammability limits for a 0.50 mm core diameter and 0.15 mm insulation thickness ETFE insulated Copper wire.

applied to the wire. In the absence of an external radiant flux the LOC of the ETFE wire was found to be 31.5%, which is in reasonable agreement with ETFE LOI of 30% [22]. The addition of an external radiant flux extended the LOC of ETFE below 31.5%. At the maximum radiant flux of the apparatus ($\dot{q}''_{ext} = 24.3 \text{ kW/m}^2$) the lowest oxygen concentration at which flame spread was observed was at 26.3%. It is seen that as the oxygen concentration is reduced, larger values of the external radiant flux are required to sustain flame propagation.

3.2. Microgravity (μg)

Figure 4 is an example of an ETFE flame propagating along the wire insulation in microgravity. Ignition of the wire insulation was preceded by the appearance of a smoke cloud that surrounded the sample near the igniter. Short flashes followed and eventually a flame wrapping around the electrical cable appeared. Flames extended along the wire in the flow direction and had a cylindrical appearance. Flames in microgravity were less bright than flames in normal gravity. Although melting of the unburned insulation occurred, there was no dripping in contrast to 1 g. The lack of dripping may cause differences between 1 g and μg flames spread. Examination of the wire samples showed that flames spreading in μg left no insulation residue deposits on the surface of the copper wire.

The experimental data for Propagation or No-Propagation as a function of the flow oxygen concentration and radiant flux applied is presented in Fig. 5. Without an external radiant flux the LOC of the ETFE wire was 26.4%, which is lower than the LOC in 1 g (31.5%). The lowest oxygen concentration at which opposed flame spread was observed was 20.3% for a radiant of 18.8 kW/m^2 . Despite the limited number of

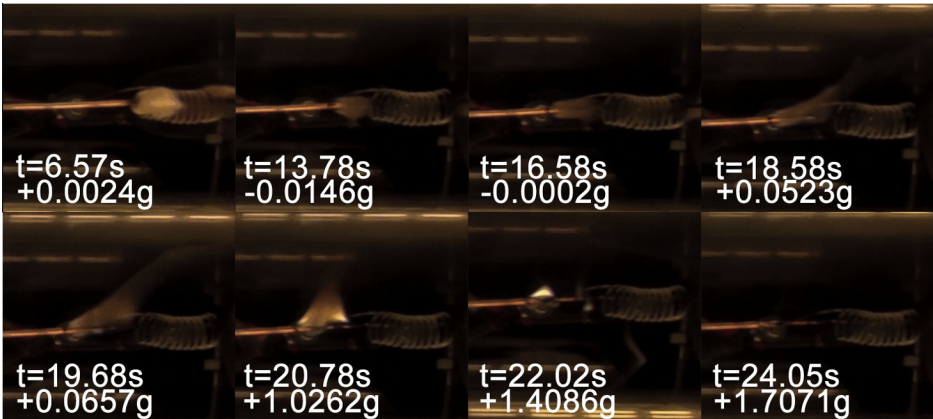


Fig. 4. ETFE combustion during microgravity conditions and subsequent extinction due to gravity. Test conditions were 21.2% O₂ and external radiant flux ($\dot{q}''_{ext}$) of 19 kW/m^2 , $t = 0.00 \text{ s}$ corresponds to the beginning of the microgravity period.

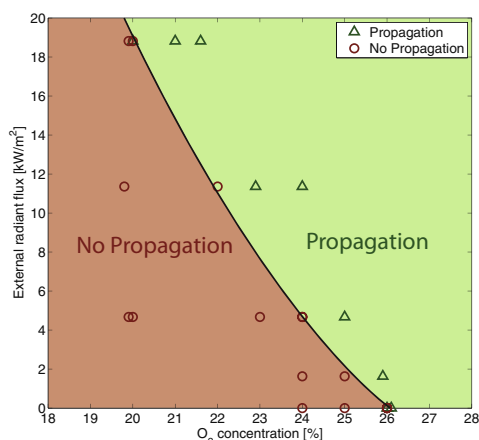


Fig. 5. Microgravity flammability limits for a 0.50 mm core diameter and 0.15 mm insulation thickness ETFE insulated Copper wire.

experiments conducted in μg , the results showed that similar to 1 g, the addition of an external radiant flux extended the flammability limits of the wire sample. The number of tests that could be performed in microgravity was small due to the limited availability of parabolic flights. However, enough data was obtained to reach conclusions about the differences in LOC between normal gravity and microgravity.

4. Discussion

4.1. LOC boundaries in 1 g and μg

The ETFE Propagation/No-Propagation boundaries in 1 g and μg are presented in Fig. 6 for comparison purposes. It is seen that both microgravity and an external radiant flux result in a decrease in the LOC of the ETFE wire. The effect of microgravity on the LOC of a material

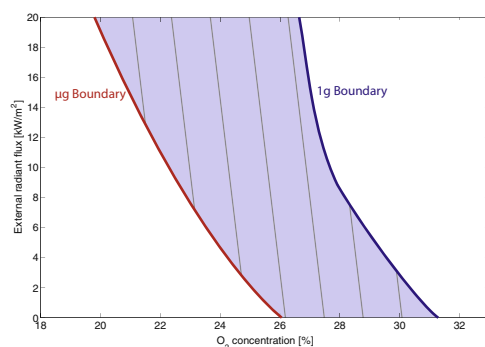


Fig. 6. Extension of ETFE insulated wires flammability boundaries in microgravity.

has been observed and discussed by other researchers [1,2,13,23]. Basically, in the absence of buoyancy and with velocity flows smaller than those induced by buoyancy in normal gravity, the flame spreads in a flow field that has a thicker boundary layer and with the flame leading edge opposing a lower flow velocity. Under these conditions the convective (and diffusive) characteristic time is larger, which allows for a larger chemical time than in normal gravity. As a consequence, it is possible to have the gas phase reaction and consequently flame spreads at lower oxygen concentrations [24]. In addition to chemical kinetics effects there are transport effects that also differentiate opposed flame spread in micro and normal gravity. These transport effects are complex and counteract each other, perhaps having a lesser overall influence that the chemical kinetics. At low flow velocities the boundary layer is thicker and consequently the flame rests further away from the surface. As a result, the heat transfer from the flame to the fuel surface is smaller. On the other hand the heat losses from the surface to the surrounding are also reduced, which counteracts the reduced heating from the flame. There are other effects related to the radiation heat transfer from the flame to the solid and surroundings [3,4], but it appears that they are less important than the chemical kinetic effects discussed above, at least for the flow velocities used in the present study.

The reduction of the LOC with the application of an external radiant flux is primarily a solid heating effect. As the oxygen concentration is reduced the flame temperature decreases and with it the heat transfer from the flame to the solid. This together with the weakening of gas phase chemical reaction deters the flame from propagating. The external radiant heating compensates for the reduction in heat transfer thus sustaining the spread of the flame to lower oxygen concentrations. Eventually the oxygen concentration is too low for the flame to spread due to the lower heating contribution from the flame and the weaker gas phase reaction. The relative effect of the external radiant flux in extending the LOC is similar in normal and microgravity.

4.2. Comparison between ETFE and PE

Given that ETFE wire is a fire resistant material and that there is no much basic information about its burning behavior, it is of interest to compare its flame spread characteristics to another wire with a better characterized insulation material. Polyethylene (PE) insulated wires have been used to study the flammability limits of electrical wires [8,10,11], thus it is a material that provides a good reference for comparison with ETFE. Takahashi et al. [10] studied the extinction limits of opposed flame spread over PE insulated wires

Table 1
LOI values for PE and ETFE insulated copper wires in 1 g and μg . Data for PE insulated wires taken from Takahashi et al. [10].

Core/Insulation	1 g LOI [% O ₂]	μg LOI [% O ₂]	ΔLOI [% O ₂]
Copper/PE	19	16	3
Copper/ETFE	31.5	26	5.5

Table 2
Thermal ignition parameters for PE and ETFE data taken from Refs. [25,22,26].

Polymer	ΔH_c (MJ/kg)	HRP [25]
PE	47.74 [26]	18
ETFE	13.7 [22]	6

over a range of opposed flow velocities. Table 1 shows the 1 g and μg flame spread limits for ETFE and PE insulated copper wires. Although there are differences in the opposed flow velocity and insulation thickness, 0.10 m/s and 0.15 mm for PE, and 0.12 m/s and 0.30 mm for ETFE, a comparison of the results provides an insight about the flame spread limits of both fire resistant and non fire resistant materials.

From Table 1 it is seen that the LOI for PE in both 1 g and μg is smaller than that of ETFE insulated wires. In addition, the difference between the 1 g and μg LOI values (ΔLOI) is smaller for PE. These differences can be explained by the differences of the thermal ignition parameters of materials [22,25,26]. Table 2 presents thermal ignition parameters for both ETFE and PE. Noticeable is the larger heat of combustion (ΔH_c) and heat release parameter (HRP) of PE. The large value of both ΔH_c and HRP may be a possible reason for the smaller ΔLOI observed in PE. In PE the relative contribution of reduced heat losses is small in comparison to ΔH_c . However, for the case of ETFE, reduction of heat losses in microgravity represents a more significant contribution.

5. Conclusion

The flame spread boundaries of ETFE insulated copper wires exposed to an external radiant flux has been studied in 1 g and μg . It has been found that the Limiting Oxygen Index (LOI) in μg is smaller than in 1 g. This reduction in the LOI of ETFE agrees with previous studies that have also shown that microgravity results in a reduction in the LOI of materials. It was also found that external radiation is able to extend the flame spread limits (LOC) to values below the LOI in normal gravity and microgravity. The magnitude of the external radiant flux required for flame spread increases as oxygen concentration decreases. This relationship between external radiant flux and oxygen concentration

can be analyzed in terms of the net heat transfer from the flame to the unburned ETFE insulation. A comparison of the ETFE and PE LOI values in 1 g and μg shows that microgravity has a more pronounced effect on the flammability of ETFE. One possible explanation for this behavior is the difference between the heat release parameter (B number) of both materials.

The results of this work are relevant given that the flammability of materials is routinely tested without considering the effects of environmental variables, which according to the results presented here, may not be indicative of the absolute flammability limits of materials.

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